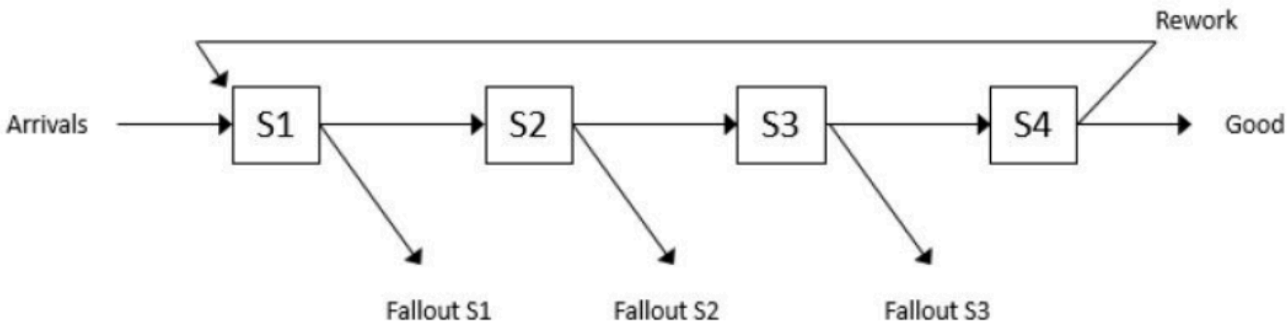
CMOR 451 HW 5

1.

(a)

performance measures:

- average number of parts in the system
- max number of parts undergoing rework in the system at the same time
- average time that parts spend in the system
- utilization of each machine S1, S2, S3, S4
- rate at which good parts are completed, in parts per hour
- rate at which parts fall out at S1, S2, S3 in parts per hour



pseudocode:

```python
cap = [1, 1, 2, 1]
service = [U(10, 20), Tria(20, 30, 40), U(60, 84), Exp(40)]
p = [0.15, 0.30, 0.33, 0.08]

class Part:
  def __init__(self, arrival, stage):
    self.rework = False
    self.arrival = arrival
    self.stage = stage
```

```
class Event:
  def __init__(self, time, type, part):
    self.time = time
    self.type = type # arrival, departure1, 2, 3, 4
    self.part = part

def simulate(T, seed):
  t = 0
  busy = [0, 0, 0, 0]
  queue = [dequeue() for _ in range(4)]
  in_system = 0
  R = 0
  maxR = 0

  area_N = 0
  area_busy = [0, 0, 0, 0]
  total_time_in_sys = 0
  completed = 0
  fallouts = [0, 0, 0]

  events = heap() # heap since insertion and pop earliest are both O(log n)

  push(events, Event(Exp(28.8), 'arrival', Part()))

  while events:
    ev = pop earliest event from event list
    if ev.time > T:
      break
    dt = ev.time - t
    area_N += in_system * dt
    for i in range(4):
      area_busy[i] += busy[i] * dt
    t = ev.time

    if event.type == 'arrival':
      push(events, Event(t + Exp(28.8), 'arrival', Part()))
      ev.part.arrival_time = t
      in_system += 1
      enter_station(0, ev.part)

    else: # departure from station i
      i = ev.station
      part = ev.part
      busy[i] -= 1
```

```
      if queue[i]:
        start_service(i, queue[i].popleft())

      if i in (0, 1, 2) and rng.random() < p[i]:
        fallouts[i] += 1
        in_system -= 1
      elif i < 3:
        enter_station(i + 1, part)
      else: # S4
        if rng.random() < p[3]:
          # rework
          R += 1
          maxR = max(maxR, R)
          part.in_rework = True
          enter_station(0, part)
        else:
          completed += 1
          in_system -= 1
          total_time_in_sys += t - part.arrival_time

def enter_station(i, part):
  if busy[i] < cap[i]:
    start_service(i, part)
  else:
    queue[i].append(part)

def start_service(i, part):
  busy[i] += 1
  push(events, Event(t + service[i](), ('departure', i), part))
```

Results:

```
Traffic equations: arrival rate into S1..S4 (parts/hr): [129.1178 109.7502  76.8251
51.4728]
                  utilization S1..S4:               [0.538  0.9146 0.7683 0.5719]

saved welch_hw5.png
Pilot: n0 = 20, warm-up = 500 min, run length = 5000 min
Measure                          Estimate                    95% CI  Half-width
Theory  Units
--------------------------------------------------------------------------------
---------------
```

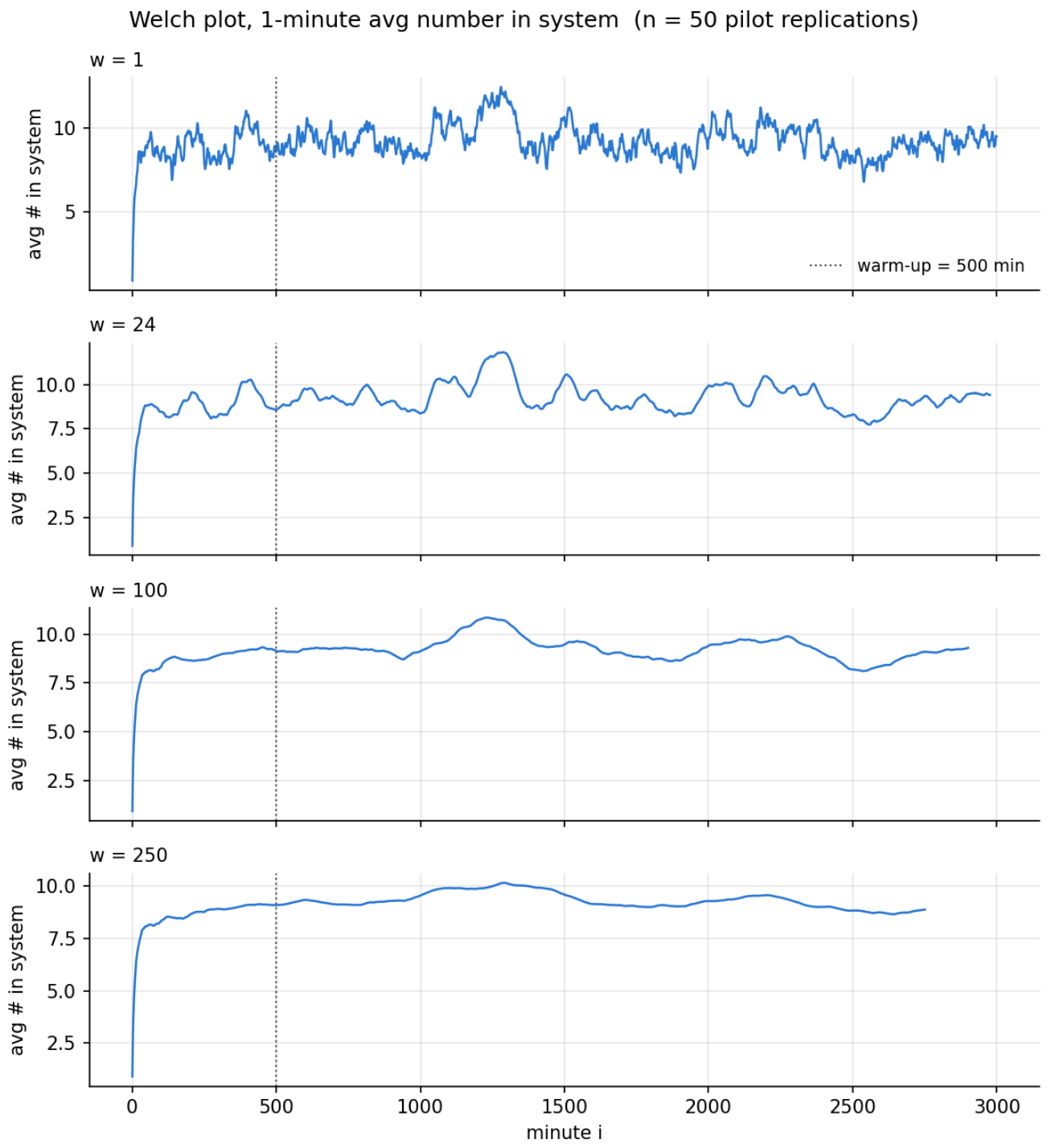
```
Avg number in system L              9.8427   [   9.3481,   10.3373]      0.4946
parts
Max parts in rework at once         4.1500   [   3.7404,    4.5596]      0.4096
parts
Avg time in system, all parts       4.7135   [   4.4895,    4.9374]      0.2240
min
Avg time in system, good parts      6.6106   [   6.3412,    6.8800]      0.2694
min
Utilization S1                      0.5388   [   0.5361,    0.5416]      0.0027
0.5380
Utilization S2                      0.9170   [   0.9121,    0.9219]      0.0049
0.9146
Utilization S3                      0.7715   [   0.7671,    0.7760]      0.0044
0.7683
Utilization S4                      0.5740   [   0.5701,    0.5780]      0.0039
0.5719
Good part rate                     47.6472   [  47.3565,   47.9379]      0.2907
47.3550  parts/hr
Fallout rate S1                    19.3512   [  19.2064,   19.4960]      0.1448
19.3677  parts/hr
Fallout rate S2                    32.8320   [  32.5777,   33.0863]      0.2543
32.9250  parts/hr
Fallout rate S3                    25.3686   [  25.0334,   25.7038]      0.3352
25.3523  parts/hr

Replications needed for +/-2% relative precision, n = (z s / (r ybar))^2:
  Avg number in system L             111
  Max parts in rework at once        428
  Avg time in system, all parts       99
  Avg time in system, good parts      73
  Utilization S1                       2
  Utilization S2                       2
  Utilization S3                       2
  Utilization S4                       3
  Good part rate                       2
  Fallout rate S1                      3
  Fallout rate S2                      3
  Fallout rate S3                      8

saved runlength_hw5.png

Production: n = 428 replications, warm-up = 500 min, run length = 5000 min
Measure                          Estimate                    95% CI  Half-width
Theory  Units
--------------------------------------------------------------------------------
```

```
---------------
Avg number in system L              9.4754   [   9.3816,    9.5692]      0.0938
parts
Max parts in rework at once         4.0794   [   4.0083,    4.1506]      0.0711
parts
Avg time in system, all parts       4.5444   [   4.5023,    4.5865]      0.0421
min
Avg time in system, good parts      6.4092   [   6.3588,    6.4596]      0.0504
min
Utilization S1                      0.5381   [   0.5376,    0.5386]      0.0005
0.5380
Utilization S2                      0.9147   [   0.9138,    0.9156]      0.0009
0.9146
Utilization S3                      0.7687   [   0.7677,    0.7696]      0.0010
0.7683
Utilization S4                      0.5714   [   0.5702,    0.5726]      0.0012
0.5719
Good part rate                     47.3577   [  47.2843,   47.4311]      0.0734
47.3550  parts/hr
Fallout rate S1                    19.3891   [  19.3456,   19.4327]      0.0436
19.3677  parts/hr
Fallout rate S2                    32.9068   [  32.8451,   32.9684]      0.0617
32.9250  parts/hr
Fallout rate S3                    25.3812   [  25.3266,   25.4358]      0.0546
25.3523  parts/hr
```



(d)

Measure	Base (d)	Reduced fallout (e)(ii)

Replications n	498	269
Warm-up / run length	500 / 5000 min	5000 / 50,000 min
L (parts)	9.528 ± 0.087	25.962 ± 0.455
Max parts in rework	4.07 ± 0.07	9.57 ± 0.18
W, all parts (min)	4.572 ± 0.039	12.459 ± 0.215
Utilization S2	0.9143 ± 0.0009	0.9753 ± 0.0004
Good parts per hour	47.34 ± 0.07	58.15 ± 0.03

Results with reduction of fallout probabilities:

(i) Lowering fallout probabilities means less parts fall out, so the later servers see more work. Utilization of S2 bumps from 0.915 to 0.975, nearly saturated. This creates a bottleneck, and average time in the system increases from 4.5 to 12.1 minutes. The run settings need updating because the 0.97 utilization needs time to warm up.

(ii)

```
Traffic equations: arrival rate into S1..S4 (parts/hr): [130.0566 117.0509  87.7882
63.2075]
                   utilization S1..S4:                [0.5419 0.9754 0.8779 0.7023]

saved welch_hw5.png
Pilot: n0 = 30, warm-up = 5000 min, run length = 50000 min
Measure                           Estimate                    95% CI  Half-width
Theory  Units
--------------------------------------------------------------------------------
---------------
Avg number in system L             27.2114   [  25.7350,   28.6879]      1.4765
parts
Max parts in rework at once        10.3000   [   9.4499,   11.1501]      0.8501
parts
Avg time in system, all parts      13.0454   [  12.3494,   13.7414]      0.6960
```

```
min
Avg time in system, good parts     15.6803   [  14.9059,   16.4547]      0.7744
min
Utilization S1                      0.5424   [   0.5417,    0.5430]      0.0006
0.5419
Utilization S2                      0.9765   [   0.9753,    0.9778]      0.0013
0.9754
Utilization S3                      0.8785   [   0.8769,    0.8800]      0.0016
0.8779
Utilization S4                      0.7012   [   0.6995,    0.7029]      0.0017
0.7023
Good part rate                     58.1776   [  58.0649,   58.2903]      0.1127
58.1509  parts/hr
Fallout rate S1                    12.9857   [  12.9522,   13.0192]      0.0335
13.0057  parts/hr
Fallout rate S2                    29.3146   [  29.2392,   29.3900]      0.0754
29.2627  parts/hr
Fallout rate S3                    24.6350   [  24.5607,   24.7092]      0.0742
24.5807  parts/hr

Replications needed for +/-2% relative precision, n = (z s / (r ybar))^2:
  Avg number in system L             203
  Max parts in rework at once        470
  Avg time in system, all parts      197
  Avg time in system, good parts     168
  Utilization S1                       1
  Utilization S2                       1
  Utilization S3                       1
  Utilization S4                       1
  Good part rate                       1
  Fallout rate S1                      1
  Fallout rate S2                      1
  Fallout rate S3                      1

saved runlength_hw5.png

Production: n = 470 replications, warm-up = 5000 min, run length = 50000 min
Measure                           Estimate                    95% CI  Half-width
Theory  Units
------------------------------------------------------------------------------
---------------
Avg number in system L             26.2211   [  25.8643,   26.5779]      0.3568
parts
Max parts in rework at once         9.7574   [   9.6056,    9.9093]      0.1519
parts
```

```
Avg time in system, all parts      12.5845   [  12.4154,   12.7537]      0.1692
min
Avg time in system, good parts     15.1694   [  14.9818,   15.3571]      0.1876
min
Utilization S1                      0.5418   [   0.5417,    0.5420]      0.0001
0.5419
Utilization S2                      0.9753   [   0.9750,    0.9755]      0.0003
0.9754
Utilization S3                      0.8775   [   0.8772,    0.8778]      0.0003
0.8779
Utilization S4                      0.7020   [   0.7016,    0.7024]      0.0004
0.7023
Good part rate                     58.1285   [  58.1044,   58.1526]      0.0241
58.1509  parts/hr
Fallout rate S1                    13.0040   [  12.9923,   13.0156]      0.0117
13.0057  parts/hr
Fallout rate S2                    29.2774   [  29.2597,   29.2950]      0.0176
29.2627  parts/hr
Fallout rate S3                    24.5750   [  24.5591,   24.5909]      0.0159
24.5807  parts/hr
```

(iii)

Measure	Mean difference	95% CI
L (parts)	+16.73	[16.38, 17.08]
W (min)	+8.03	[7.86, 8.19]
Good parts per hour	+10.79	[10.78, 10.80]